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Abstract

Phylogentic trees are often the first step in many evolutionary analyses. The *phangorn* package has become one of the most important packages to infer phylogentic trees in R. *phangorn* contains a variety of functions to infer phylogenetic trees from distances, with maximum parsimony or with maximum likelihood. It contains functions for reconstructing ancestral sequences, tree comparison and to visualize conflicting signals. We describe some significant features of and recent updates to the *phangorn* R package and use the opportunity to illustrate several popular workflows of the software.

Key words: phylogentics, exploratory data analysis, reproducible research

Introduction

A phylogenetic tree shows relationship among evolutionary related objects in form of a connected graph without cycles. Objects are nowadays most often biological sequences, but can be also morphological measurements, languages, manuscripts.

The development of phylogenetics has been tightly linked to the use of computers since the late 1950's (Felsenstein 2004). During many decades, the computations needed for phylogenetic inference were done with closed programs. The rise of free and open-source software in the late 1990's stimulated the development of a radically new approach based on the open collaborations and code sharing through the adoption of general public licencing. R, which was created in 1993, appeared an ideal choice to implement phylogenetic methods in the philosophy of sharing and collaborating.

When the development of *phangorn* started no phylogentic inference methods were available in R apart from distance based methods. *phangorn* introduced the possibility to infer phylogenies with Maximum Parsimony or Maximum Likelihood directly in R. Felsenstein (2004) and Yang (2014) offer more background, and mathematical detail over the methods we describe. A strength of R is that there are the capabilities to manipulate data and visualization. This allows to perform exploratory analysis (Tukey et al. 1977) which otherwise needs several different programs.

Additionally R offers a huge ecosystem devoted to phylogenetic comparative methods which make use of phylogenies like *phytools* (Liam J. Revell 2012; Liam J. Revell 2024), *geiger* (Pennell et al. 2014), *ape* (Emmanuel Paradis and Schliep 2019) or *vegan* (Oksanen et al. 2022) and many more.

Nowadays there exist several R packages which allow to call specialized phylogenetic inference software like *babette* (Bilderbeek and Etienne 2018) to BEAST2 (R. Bouckaert et al. 2014), *Revticulate* (Charpentier and Wright 2022) to *RevBayes* (Höhna et al. 2016) or *Rphylip* (Liam J. Revell and Chamberlain 2014) to phylip (Felsenstein 2013) and *ips* (Heibl, Cusimano, and Krah 2019) to several programs like on *RAxML* (Stamatakis 2014) and *MrBayes* (Ronquist and Huelsenbeck 2003).

The original article describing *phangorn* (K. P. Schliep 2011) was written over a decade ago and is in need of an update. With this article we want to take the opportunity to not only offer a placeholder for new citations, but take the opportunity to showcase some common workflows, several we improved and streamlined recently.

Phylogenetic reconstruction

The phyDat format

First we want to use the opportunity to introduce the **phyDat** object for storing alignments. Aligned sequences have a matrix form (see table 1). The **phyDat** format is similar to a **factor** and allows to store categorial data. Whereas in a **data.frame** each column or variable might have different categories, an alignment shares all the categories for all positions. Currently *phangorn* differentiates between four types of data, "DNA" to store nucleotide data, "AA" for amino acids, "CODON" for codon data and finally "USER" for any kind of discrete / categorial data. The amino acid models are starting to get with substitution matrix for structural phylogenetics (Kempen et al. 2024; Garg and Hochberg 2025; Moi et al. 2025; Puente-Lelievre et al. 2023; Puente-Lelievre, Malik, and Douglas 2025).

There are convenience functions **dna2codon** and **dna2aa** to translate nucleotides to codons or amino acids.
With the help of a contrast matrix we define ambiguous states (Felsenstein 2004). Most phylogenetic software code gaps, usually coded by "-", as ambiguous states assuming characters might not have been sequenced. On the other hand one might want to treat gaps as a state on its own right resulting from an insertion or deletions. *phangorn* provides a convenience function **gap_as_state** to switch easily from coding gaps as ambiguous to coding gap as a state for either nucleotide sequences or amino acids.

Now we start with a parsimony analysis using categorical morphological data. While it is most common nowadays to construct phylogenies from nucleotide sequences or amino acids, the data format in *phangorn* is quite flexible allowing any kind of discrete data.

Parsimony example with morphological data

A hurdle for morphological analysis is reading in the data and afterwards coding these properly. Here we profit from the data processing capabilities of the R environment. The data set we are using contains morphological data for 12 mite species (Schäffer et al. 2010), with 79 encoded characters, see table 1 for a subset of the date. When reading in the *.csv* file, **row.names = 1** uses the first column (species) as row names. To get a **phyDat** object we are using for the analysis, we have to convert the data.frame into a matrix with **as.matrix**.

```
library(phangorn)
fdir <- system.file("extdata", package="phangorn")
mm <- read.csv(file.path(fdir, "mites.csv"), row.names=1)
mm_pd <- phyDat(as.matrix(mm), type="USER", levels=0:7)
```

Table 1: Excerpt from the original mites data set showing the first 6 species and 8 variables. See main text for additional details.

	X1	X2	X3	X4	X5	X6	X7	X8
S._alpinus	2	2	3	2	4	4	0	2
S._arenocolus	2	2	3	2	4	7	0	2
S._ianus	2	2	3	2	4	7	0	2
S._minutus	2	2	3	2	4	7	0	2
S._pannonicus	2	2	3	2	4	7	0	2
S._pictus	3	2	1	0	4	6	0	2

Now that we have our data, we can start the analyses. Maximum parsimony (MP) tries to find the tree with the least number of substitutions explaining the data. We use the function **pratchet** for a heuristic search of the tree space. It implements the parsimony ratchet (Nixon 1999) and uses subtree pruning and rearrangements (SPR) as a tree rearrangement.

```
mm_tr <- pratchet(mm_pd, minit=1000, maxit=5000, all=TRUE)
mm_tr <- acctran(mm_tr, mm_pd) |> midpoint()
plot(ladderize(mm_tr[[1]]), show.node.label=TRUE, cex=0.6)
add.scale.bar(y=1, x=0, cex=0.6)
```

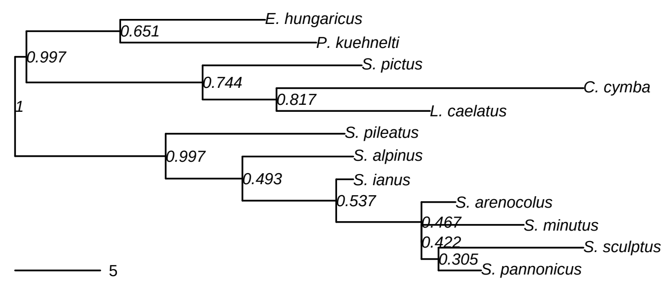

Figure 1: One of the maximum parsimony trees with bootstrap support and edge length assign using ACCTRAN. See main text for additional details.

Here we specified a few additional arguments. In this manuscript we frequently will set the argument **trace=0** or **pml.control(trace=0)**. This is done to suppress printing out progress of the current parameter estimates, so that the output in the document and insider R are as close as possible. With **all=TRUE** we get all trees with lowest parsimony score which were visited during the search in a **multiPhylo** object. The **minit** and **maxit** set the number of iterations for the ratchet. Since we set a minimum of 1000 iterations, we already have at least 1000 bootstrap samples for edge support as a byproduct of the parsimony ratchet. For larger trees this might takes some time and one might want to reduce the number of iterations. The trees returned by **pratchet** contain no edge weights. At last we assign edge lengths, but have to keep in mind that often there will be not a unique way to assign edge lengths.

The first of the most parsimonious trees is shown in figure 1 with bootstrap values, which are stored at the nodes. Finally we can export these trees using the *ape* function **write.tree** to save in Newick or **write.nexus** to store in Nexus format (Maddison, Swofford, and Maddison 1997).

Branch and bound

In the case of our mites-dataset with 12 sequences, it's also possible to apply a branch and bound algorithm (Hendy and Penny 1982). This algorithm guarantees to find all most parsimonious trees, but in the worst case the algorithm has to evaluate every tree and it can be very time consuming. If characters are supporting a specific tree, especially if characters are homoplasy free, this can be very effective. With bigger datasets (more than 20 taxonomic units) it is definitely recommended to use **pratchet**. The following lines of code have been used to compute the trees using branch and bound, compute a consensus network and plot it a splits graph (Dress and Huson 2004) as shown in figure 2.

```
mm_bab <- bab(mm_pd)
cnet <- consensusNet(mm_bab, p=0.2)
plot(cnet, show.edge=TRUE, col.edge="red", edge.width=1.5,
     edge.color="grey", cex=0.5, use.edge.length=FALSE)
```

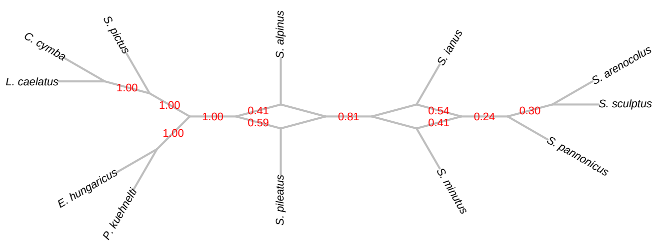

Figure 2: Consensus network of the 37 most parsimonious trees. See main text for additional details.

In this case we found even one tree more than in the original article (Schäffer et al. 2010). We build a consensus network (Holland et al. 2004) containing all splits which are shared in at least 20% (eight trees) of all most parsimonious trees.

The package *tanggle* (K. Schliep et al. 2022) provides an alternative to plot split graphs and explicit networks extending the *ggtree* package (Yu et al. 2017) in an "grammar of graphics" frame work (L. Wilkinson 1999; Wickham 2016).

Additionally the functions `read.nexus.networx` and `write.nexus.networx` can import and export split networks as nexus files together with some annotation to exchange with *Splitstree* (Huson and Bryant 2006) and other software.

Classical Maximum likelihood analysis

For molecular sequence data maximum likelihood (ML) and Bayesian MCMC inference is nowadays more common than maximum parsimony. Maximum likelihood inference for phylogentic trees offers a full statistical framework and estimates branch length, tree topology and substitution parameters (Felsenstein 1981, 2004). For large trees (> 1000 taxa) *iqtree* (Nguyen et al. 2015) or *RAxML* (Stamatakis 2014) might be considerably faster.

As input we need aligned nucleotide or amino acid sequences and we can read in these directly with several popular file formats like fasta, phylip or nexus. In case of DNA or AA data there is no need for a transformation as in the example above.

```
dna <- read.phyDat("Laurasiatherian.fas", format="fasta")
```

Additionally the function `as.phyDat` can be used to transform several common used formats in *ape* or *bioconductor*.

As a first we select the best fitting transition model. For this we call the function `modelTest` to compare different nucleotide or protein models. This is similar to popular programs ModelTest, ProtTest or ModelFinder (D. Posada and Crandall 1998; David Posada 2008; Abascal, Zardoya, and Posada 2005; Kalyaanamoorthy et al. 2017). `modelTest` is one of the functions which support parallel computing. The other functions are `bootstrap.pml`, `bootstrap.phyDat` and `superTree`. *phangorn* now uses the future package (Bengtsson 2021), which allows to use different backends for the users to choose depending on their hardware and operating system. For more information consult the documentation of the future package. By default all available nucleotide or amino acid models are compared, but one can restrict the models to be tested by providing a vector of models to test. Furthermore to speed up computations restrict the models to be estinmated. In a first step all models are estimated without any model of rate heterogeneity among sites (RHAS). Next only for the best fitting models of these the models of RHAS are estimated. Afterwards we compute the most promising models.

```
mt <- modelTest(dna, model=c("JC", "K80", "HKY", "GTR"),
                RHAS=c("gamma", "gamma_weighted", "free_rate"))
```

Table 2: The 10 best models according to the BIC and subset of columns.

Model	df	logLik	AIC	BIC	TL
GTR+R(4)+I	106	-44437.67	89087.34	89730.16	5.034043
GTR+GW(4)	103	-44473.09	89152.19	89776.81	4.760176
GTR+R(4)	105	-44471.12	89152.24	89788.99	4.786844
GTR+GW(4)+I	104	-44492.67	89193.35	89824.04	4.858513
GTR+G(4)+I	101	-44596.21	89394.42	90006.91	5.675592
GTR+G(4)	100	-44728.70	89657.39	90263.83	6.196003
HKY+R(4)+I	102	-44765.34	89734.68	90353.24	5.412475
HKY+R(4)	101	-44793.28	89788.56	90401.06	5.274061
HKY+GW(4)	99	-44806.11	89810.21	90410.58	5.019978
HKY+GW(4)+I	100	-44836.60	89873.20	90479.63	5.059369

The functions returns a matrix with common goodness-of-fit measures the AIC, AICc or BIC. And additionally some parameter estimates like the shape parameter of the (discrete) gamma distribution (Yang 1994; Waddell 1995), the proportion of invariant sites and the total tree length are returned in the matrix. Table 2 shows the best 10 models according to the BIC (Schwarz 1978). In this case the best of the tested models according to the BIC is GTR with invariant sites and a free rate model accounting for rate variation.

Depending on the data set there can be quite striking correlations between different parameters. For example in **?@fig-ggrepel** and table 2, we can see that the choice of the model for transition site variation have a strong influence on the total tree length for this data set. For other data sets we might observe a strong correlation between the transition transversion ratio and the tree length. So model choice should not only rely solely on a single goodness-of-fit measures, but also use exploratory data analysis (Tukey et al. 1977; Liam J. Revell et al. 2018), to avoid possible biases (Jia 2014; Ferretti et al. 2025).

The object returned by the function `modelTest` also the list of estimated trees and the call object to reconstruct each model with the function `as.pml()`. The function `modelTest` not only returns the table of goodness-of-fit measures, but also stores the associated trees and model parameter which can used to head start the analysis. These are saved as a `call` together with the optimized trees in an environment and the function `as.pml` evaluates this call to get a `pml` object back to use for further optimization or analysis. This can either be done for a specific model (e.g. `as.pml(mt, "BIC")`), or for a specific criterion (e.g. `as.pml(mt, "HKY+G(4)+I")`).

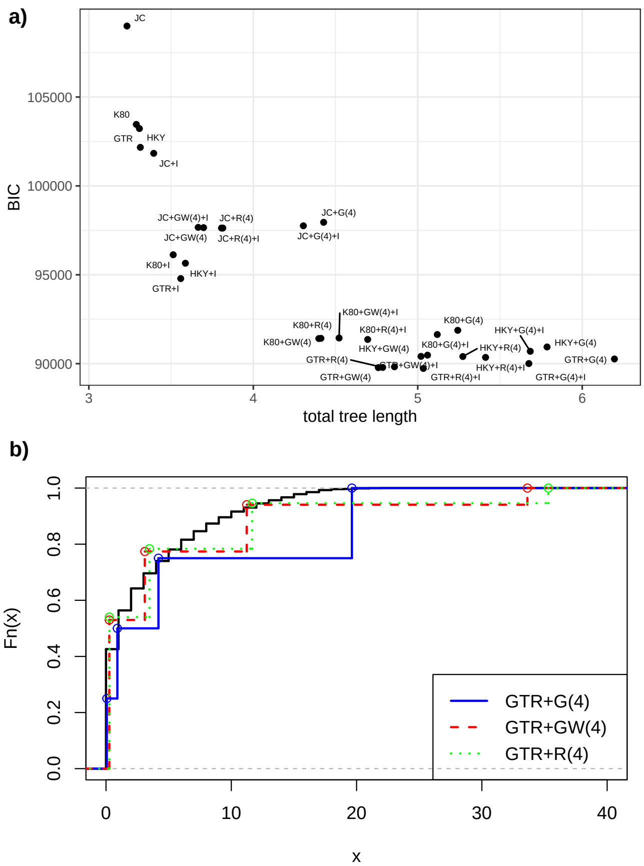

Figure 3: a) Scatter plot showing the log-likelihood and total tree length for the different models of the modelTest. The total tree length differs considerably between the models. b) Cumulative distribution function for the observed changes per site and the several RHAS. See main text for additional details.

The following code allows us to compare different RHAS models and see how close these fit the data.

```
plotRates(as.pml(mt, "GTR+G(4)"), xlim = c(0, 40))
plotRates(as.pml(mt, "GTR+GW(4)"), append = TRUE,
          cdf.color = "red", lty = 2)
plotRates(as.pml(mt, "GTR+R(4)"), append = TRUE,
          cdf.color = "green", lty = 3)
```

Figure 3 b) shows the cumulative distribution function for the observed number of substitution per site, which is the parsimony score for each site pattern for the tree used in modelTest. We compare this distribution with the different models of rate heterogeneity describing visually how close these models are to the data. To achieve this we need to multiply the rate of each class with the total tree length of each tree. In this case the freerate and discretized gamma model with variable weights behave very similar in their steps and the total tree length.

After we found the best fitting model we can start optimizing all parameters including performing tree rearrangements. For this we use the function `pml_bb`, which internally calls the `optim.pml`. The function `pml_bb` has one mandatory argument `x`, but this input can have different classes:

1. An alignment, that is an object of class `phyDat`, or from `AAbin`, `DNAbin` from the *ape* package. In this case we need to provide also a model term e.g. "HKY+G(4)". Additionally one can also supply a starting tree. If no tree is supplied than a starting tree is computed.
2. An object of class `modelTest` and `pml_bb` will extract the model parameters according to the best model according to the BIC (AICc or AIC). So the optimization starts with already some parameters optimized.
3. An object of class `pml`. This can be the output of analysis from 1. or 2. It is often a good idea to estimate a tree using NNI rearrangements before doing extensive topology search using "stochastic" or "ratchet". So one can have a look on a reasonable tree and also can get a feeling on long how the full analysis might take. The functions `pml.control` and `ratchet.control` allow to control the thresholds and iterations for the optimization.

Next we showcase `pml_bb` providing a `modelTest` object we computed before.

```
(fit_mt <- pml_bb(mt))
```

```
model: GTR+R(4)+I
loglikelihood: -44405.18
unconstrained loglikelihood: -17300.92
Total tree length: 5.056478
    (expected number of substituions per site)
Minimal tree length: 3.075495
    (observed substitutions per site)
Proportion of invariant sites: 0.3199995
Model of rate heterogeneity: Free rate model
Number of rate categories: 4
        Rate Proportion
1  0.0000000 0.31999948
2  0.2481187 0.35636489
3  1.3989255 0.21212078
4  4.1673141 0.09641172
5 14.1070113 0.01510314

Rates:
a <-> c : 3.904179
a <-> g : 16.68754
a <-> t : 4.320667
c <-> g : 0.4644116
c <-> t : 29.23377
g <-> t : 1

Base frequencies:
        a         c         g         t
0.3321866 0.1990791 0.2040652 0.2646691
```

There is often some confusion about what the units of the edge lengths in phylogenetic tree represent. The total edge lengths which are reported for the maximum likelihood tree are the expected number of changes per site (like). The minimal tree length is the parsimony score divided by the number of sites. For parsimony trees (e.g. figure 1 the (total) the edge length number of substitutions on this site, i.e. not divided by the number of sites. With tip dated trees we can distinguish the (mutation) rate and time and edge length in figure 5 are there proportional to time in years. The expected number of substitutions is therefor the product of the rate and edge length and in the example below roughly around 0.0025 mutations per site per year. We know from theory that the expected number of substitutions should be larger than the observed number of substitutions.

These values are an aggregating over all sites and substitutions. There can be huge differences between different substitutions (e.g. a high transition transversion ratio) and between different sites (e.g. rate variation models) or parts of the tree (covarion models). And it is the case. When the expected number of substitutions is considerably higher than the parsimony this might need further investigation. It could be a sign of saturation and one has to be cautious for the interpretation of edge length.

The function `pml_bb` here takes an object returned by `modelTest` and performs now additional to optimizing the parameters of the model also tree rearrangements. At the time of writing there are three options for the argument `rearrangements` available: Nearest neighbor interchange (`"NNI"`) performs local tree rearrangements. This is usually pretty fast but only guarantees to find a local optimum. The option `"stochastic"` implements stochastic rearrangement algorithm following in (Nguyen et al. 2015) and `"ratchet"` the ML ratchet (Vos 2003; Nixon 1999). This might take quite a substantial amount of time, but often will result in better trees as they are able to escape local optima. And as a byproduct both these options return a bootstrap sample. With the default case of `rearrangements="stochastic"` "ultrafast bootstrapping" (Minh, Nguyen, and Haeseler 2013) and for `rearrangements="ratchet"` the classic bootstrap (Felsenstein 1985; Penny and Hendy 1985, 1986) are returned in the slot `bs` as a by-product. As an alternative we can explicitly run an bootstrap analysis after using the function `bootstrap.pml`.

We can also have a look at the phylogeny using the generic function `plot` (see figure 4 a)). A generic function is a function whose behavior depends on the class of the input object. In this case `plot` calls the the function `plot.pml`. The function `pml.pml` is just a convenience function based on `plot.phylo` function from the *ape* package.

```
plot(fit_mt, cex=0.5, method=c("FBP", "TBE"), main="a)")
terraces(fit_mt, dist_fun = "KF.dist", main="b)")
```

In figure 4 the edge length are the expected number of substitution per site, as we cannot distinguish between mutation rate and time. If an bootstrap analysis was implicitly performed support values are shown. Here the classical Felsenstein bootstrap proportions as well the transfer bootstrap (Lemoine et al. 2018) are shown next to the edges.

We additionally look at how closely bootstrap trees and the ML tree are related using the function `terraces` (Sanderson, McMahon, and Steel 2011; Sanderson et al. 2015), see figure 4 b). In this plot the first two dimensions shows the projection of pairwise tree distances using multidimensional scaling (MDS)(Gower 1966) the third dimension the log-likelihood of the trees. Here the pairwise distances were computed using the Kuhner-Felsenstein distance (Kuhner and Felsenstein 1994), but one can provide any other distance functions.

phangorn contains several functions to compute distances between trees. The symmetric or Robinson-Foulds distance (Robinson and Foulds 1979, 1981) and its weighted version, the Kuhner-Felsenstein

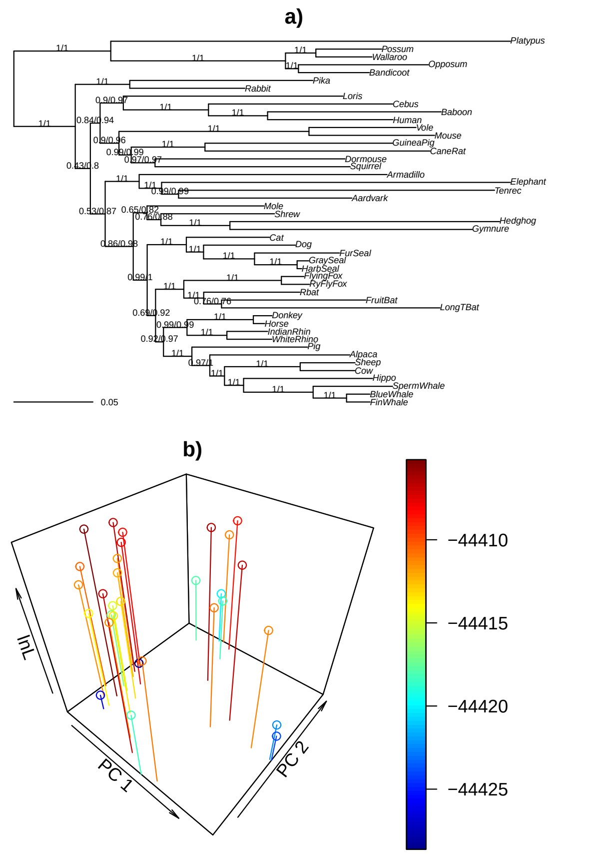

Figure 4: a) ML tree with approximate bootstrap support. The unrooted tree is midpoint rooted. b) Terrace plot for the trees based on the bootstrap samples. See main text for additional details.

distance (Kuhner and Felsenstein 1994), the path distance (Steel and Penny 1993) and the approximate SPR - distance (Oliveira Martins 2008; De Oliveira Martins, Mallo, and Posada 2016).

This complements several other R packages which also offer tree distance measures distances e.g. quartet distance (Estabrook, McMorris, and Meacham 1985) in the package Quartet (Smith 2019), geodesic distances (Billera, Holmes, and Vogtmann 2001) in the package *distory* (Chakerian and Holmes 2020) and *TreeDist* (Smith 2020, 2022).

A `pml` object can get exported using the function `write.pml`. This generates several files: the alignment of the original data, the ML tree in an newick file, the bootstrap trees if available as nexus files, a text file with the parameter estimates and an `*.rds` file containing the `pml` object. This also allows to regenerate the objects in a future R session for example for further optimization or ancestral reconstruction etc. Furthermore it exports plots of the phylogeny, terraces and consensus trees if bootstrap analysis was performed. Additionally the function generates plots similar to figure 4.

Inference of rooted trees

Zuckerkandl and Pauling proposed the molecular clock in the 1960s (Zuckerkandl and Pauling 1965). Most programs for ML inference return unrooted trees. Rooted phylogenies can tell us about the ordering of the divergence times. *phangorn* offers the estimate an ultrametric trees (Emmanuel Paradis et al. 2023), that means all tips have the same distance to the root. In this case we still would need information about some internal nodes from fossils record to separate time and mutation rates. For fast evolving organisms or viruses we can estimate mutation rates and divergence times without the need of fossil calibration. In this case we only need molecular sequences and their associated sampling times. Most implementations for maximum likelihood analysis return unrooted trees. The `pml_bb` function offers the opportunity to estimate also ultrametric and tipdated phylogenies under a strict clock. The only other ML program we are aware of inferring tipdated phylogenies is *treetime* (Sagulenko, Puller, and Neher 2018). However *treetime* is very limited when it comes to tree rearrangement. There are several Bayesian phylogenetic tools (e.g. *BEAST* (A. J. Drummond and Rambaut 2007)) which allow to estimate rooted phylogenies and additional do not rely on a strict molecular clock, but use relaxed clock models.

Inference of tipdated phylogenies

To fit a tipdated phylogeny we import an alignment and additionally a table containing the labels and the sampling date of the sequences. We than transform the two columns to an named vector to enable subsetting and ensure that the dates match.

```
tmp <- read.csv("data/H3N2.csv")
H3N2 <- read.phyDat("data/H3N2.fas", format="fasta")
dates <- setNames(tmp$numdate_given, tmp$accession)
head(dates)
```

```
KF789866 CY148382 GQ895004 CY115546 CY001279 CY009150
2013.405 2012.838 2009.482 2009.523 2000.134 2000.682
```

phangorn can estimate tipdated phylogenies assuming a strict clock using from distances and using maximum likelihood. First we infer a tree with sUPGMA (A. Drummond and Rodrigo 2000), which extends the ultra-metric method UPGMA (Sokal and Michener 1958) to tipdated data.

```
dm <- dist.ml(H3N2, "F81")
tree_supgma <- supgma(dm, dates)
```

`supgma` takes as input a distance matrix and the vector with the tipdates and return a tree, where the tip dates are constraint. Next we estimate the tipdated phylogeny using maximum likelihood. For this we use again the function `pml_bb`, but need to specify the argument method as “tipdated” and provide the tip dates. A warning at the start, the optimization of a rooted phylogeny takes considerable more time than for an unrooted phylogeny.

```
(fit_td <- pml_bb(H3N2, model="HKY+I" , method="tipdated",
                  tip.dates=dates) )
```

```
model: HKY+I
loglikelihood: -3117.85
unconstrained loglikelihood: -2883.911
Total tree length: 0.1321664
    (expected number of substituions per site)
Minimal tree length: 0.1279318
    (observed substitutions per site)
Proportion of invariant sites: 0.6865984
```

```
Rate: 0.002553314

Rates:
a <-> c : 1
a <-> g : 9.866806
a <-> t : 1
c <-> g : 1
c <-> t : 9.866806
g <-> t : 1

Base frequencies:
        a         c         g         t
0.3097759 0.1928617 0.2376819 0.2596805

Rate: 0.002553314
```

If no starting tree is supplied `pml_bb` first estimates a tipdated phylogeny based on sUPGMA (A. Drummond and Rodrigo 2000). A major difference between tip dated, ultrametric and unrooted trees is that the number of parameter to estimate edge weights differs. For unrooted trees we have one parameter to estimate the length of each edge. In case of an binary tree this is $2n-3$, where n is the number of tips. For ultrametric trees the number of internal nodes, for a binary tree this is $n-1$. In case of tipdated phylogenies with a strict clock we have one additional parameter for the mutation rate.

In figure 5 we highlight different ways to visualize uncertainty in the topology and in divergence times. In figure 5 a) we use the function `DensiTree` to plot two trees on top of each other. This is a way to show few trees in the same plot. The classical use of `DensiTree` to describe the distribution of trees and its speciation times is shown in panel c). Figure 5 a) gives a summary of describing the bootstrap tree is presented in form of support values and box plots are used to represent the divergence times.

```
densiTree(c(tree_supgma, fit_td$tree), width=2, cex=.6,
      alpha=1, col=c("blue", "red"), tip.dates=dates,
      jitter=list(amount=.2, random=FALSE),
      ylim=c(-3, 20), main="a)", direction="down")
legend("topleft", c("sUPGMA", "ML"), lty=1, lwd=2,
       bty="n", col=c("blue", "red"))
plot(fit_td, align.tip.label=TRUE, main="b)", cex=.6,
      direction="down", y.lim=c(-3, 20), x.lim=c(0,20)) |>
   add_boxplot(cex=2, outline = FALSE)
densiTree(fit_td, width=2, tip.dates=dates, main="c)",
      direction="down", ylim=c(-3, 20), cex=0.6)
```

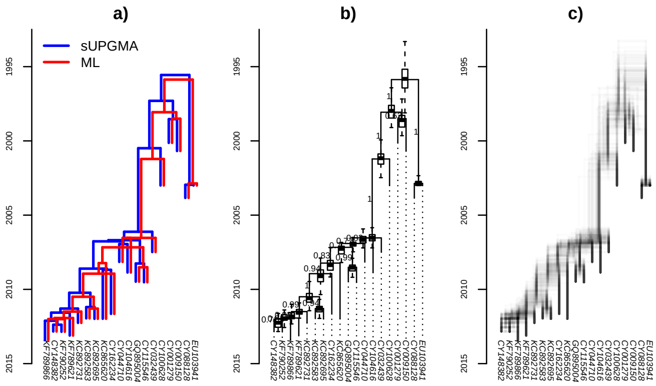

Figure 5: a) Tipdated sUPGMA (blue) and maximum likelihood tree (red) asuming strict clock model. b) ML tree with bootstrap support and uncertainty of the divergence times represented by boxplots. c) DensiTree for the same data. See main text for additional details.

An alternative representation is a densiTree (R. R. Bouckaert 2010) in figure 5 c). A nice property of this approach is that we present all the information available. All trees are plotted on top of each other, whereas otherwise a summary of this information is presented in form of support values and a box-and whisker plots for the divergence times.

Currently there are three flavors of support values available. Support values can be based on splits in case of unrooted trees (classical bootstrap), on clades for rooted trees and transfer bootstrap (Lemoine et al. 2018). The transfer bootstrap might be more appropriate for phylogenies with many taxa (Zaharias, Lemoine, and Gascuel 2023). In case of these small dataset all support values are often very similar.

Ancestral sequence reconstruction

Often we are not only interested in the tree (topology) but also in the ancestral sequence reconstruction (ASR) for specific nodes. ASR is becoming a common tool in biotechnology (Spence et al. 2021; Nicoll et al. 2023; Thomson, Carrera-Pacheco, and Gillam 2022; Prakinee et al. 2024). For example ASR of enzymes are often used as a starting point to engineer more heat tolerant enzymes or improve the reactions van der Pol, Elske et al. (2026).

We roughly follow the analysis from Sun, Calderini, and Kourist (2021). We first will load in an mafft alignment of 12 amino acids and clean up that alignment. Furthermore we assume that gaps are a state represent absence of any nucleotide and not an ambiguous state.

```
aa <- read.phyDat("data/FAP_mafft.fas",
                  format="fasta", type="AA")
aa <- gap_as_state(aa)
```

In the following we will infer the ASR with maximum likelihood, but *phangorn* additionally offers the possibility to perform ASR with maximum parsimony. We first infer the ML tree and the associated parameter. In absence of knowledge of an out group we midpoint root the tree.

```
fit <- pml_bb(aa, model="WAG+GW(4)", rearrangement="NNI")
fit <- update(fit, tree=midpoint(fit$tree))
anc_ml <- anc_pml(fit)
```

An object of class **ancestral** contains several slots: the original alignment and the phylogenetic tree with node labels the ancestral reconstruction is based on. Additionally it contains an object with probabilities of each state for each internal node and site as the result of a marginal reconstruction. And the most likely state for each node as the result of a joint reconstruction (Pupko et al. 2000). When a model with rate variation is used the most likely state based on the marginal reconstruction is used. These objects can be exported in a format similar to *iqtree* and read in again to plot the reconstruction (see figure 6).

In phangorn several plot options are to visualize ancestral reconstructions, see figure 6.

Figure 6 a) shows the phylogenetic tree with the proportion of each state shown as a pie diagram for each node with the marginal ancestral reconstruction for the 410th site (column) of the alignment. Additionally the amino acid transitions are added to the most likely positions in the tree.

```
plotAnc(anc_ml, 410, scheme="Clustal", pos="right",
        x.lim=c(0, 4.5))
add_mutations(anc_ml, pos=410, adj=c(0.5, -0.3), cex=.8)
anc_heatmap(anc_ml, scheme="Clustal", select=400:440,
            show.aa=TRUE, cex=0.6, aa.cex=0.6, legend=FALSE)
```

Fig-ancestral b) eases to comparison of ancestral and tip states in form of a heatmap as these rows close to each other.

Exploring trees

The result of a phylogentic analysis is often a sample of trees, not just a single tree as we have seen already above. For example the samples of trees can represent gene trees, trees from a bootstrap or MCMC sample. *phangorn* in combination with *ape* offer several functions to process samples of trees and exploring these highlighting agreement and differences between them.

Processing trees

Before we run tree comparisons we often need to process the trees. Many comparison are based on support values, but we have to remember that support values are properties of splits and clades, even though support values stored at node or edge labels. There can be several edges which code for the same split, e.g. edges connecting a node of degree 2 or in more general graph structures like in splits networks as in figure 2.

- To root the trees with an out-group or at an node the *ape* function **root** is very flexible. Midpoint rooting is available through the function **midpoint**, if we just want to have a nicer looking tree. (Czech, Huerta-Cepas, and Stamatakis 2017) highlighted that re-rooting might lead to wrong assignment of support values if these values are stored at the nodes. The function **root** in *ape* and **midpoint** can take care of this. However if the we support values on rooted trees which are based on clade probabilities, we always need to reassign support values as these might have changed.

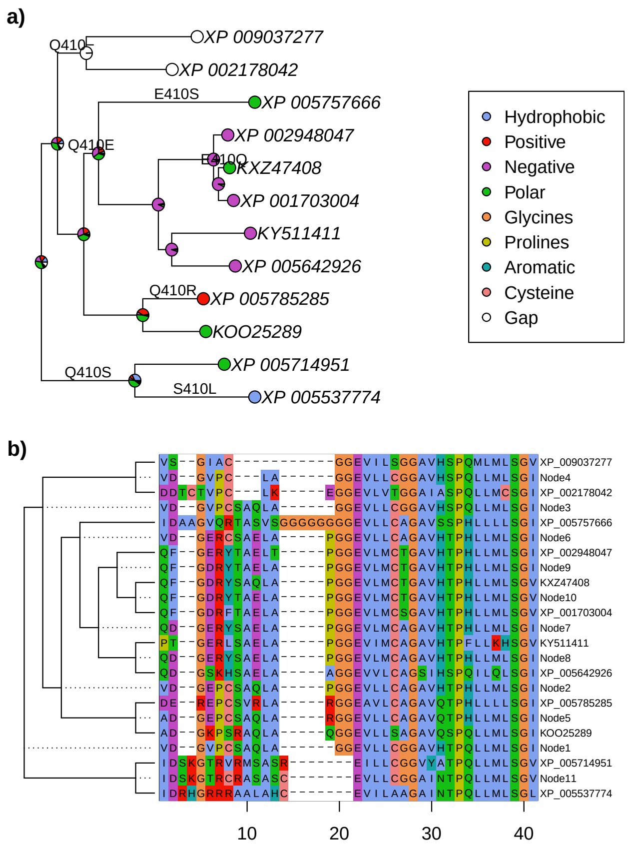

Figure 6: a) Maximum likelihood tree with the marginal reconstruction for the 410th character shown at each node. The mutations for this site are shown in their most likely position based on the joint reconstruction. b) Heat map for amino acid positions ranging from 400 to 440 in the alignment, together with joint reconstruction. See main text for additional details.

- Another transformation is pruning trees to a certain set of tips. For example we might only show the tree for which we have data collected for phylogentic comparative methods. *ape* provides the generic functions function `drop.tip` and `keep.tip` to prune the tree. It is important to perform pruning before adding support values as these values will likely change and in case of so-called "rogue" taxa support values will increase dramatically.
- Remove short edges close to zero: Short edges can lead to spurious results for support values. Therefor we want remove edges which are close to zero. Many phylogenetic tree reconstruction methods return binary trees, even though some edges are zero or close to zero so a multifurcation is more appropriate. By default the shortest branch length returned by `pml_bb` is set by default is 1.0×10^{-8}. This default can be changed with the argument control and the helper function `pml.control`. To make things worse some phylogenetic reconstruction methods might depends on the input order of taxa, so that we would always see only one site pattern. To avoid this problem we can use the function `di2multi`. If the trees are ultrametric or tipdated setting the argument `tip2root = TRUE`, ensures that is also the case after removing the edges. So for rooted bootstrap trees we use a line like `di2multi(tree, tol=1e-7, tip2root = TRUE)`.

All the functions (`root`, `midpoint`, `drop.tip`, `keep.tip`, `di2multi`) mentioned in the paragraph above are generic. This means so they will work on a single tree, but also list of trees, an object of class `multiPhylo`. To promote reproducible research and support FAIR principle (M. D. Wilkinson et al. 2016; Jacobsen et al. 2020) it should be best practice to provide bootstrap or MCMC sample from the analysis and not only supplying a "best" tree with associated information like support values. This allows transform the trees (re-rooting, pruning) and assign afterwards support values.

In the following we taken 100 trees from a MCMC sample from (Upham, Esselstyn, and Jetz 2019) and prune these trees down to 25 tips.

```
trees <- read.tree("data/trees/Upham_trees_100.tre")
trees <- .compressTipLabel(trees)
short_nam <- attr(trees, "TipLabel") |> abbreviateGenus()
attr(trees, "TipLabel") <- short_nam
keep <- sample(short_nam, 25) # sample 40 names
trees <- keep.tip(trees, keep)
trees <- di2multi(trees, tol=0.5, tip2root = TRUE)
```

Consensus trees and networks

The next task is to compute a consensus tree. *Ape* provides the function `consensus` to compute strict (argument p=1) and majority consensus trees (argument p=0.5). These consensus trees might not be binary, but contain multifurcations. *Phangorn* extends these with the functions `maxCladeCred`, `allCompat` and `consensusNet`, which resolves multifurcations and in case of `consensusNet` can show alternative splits:

- allCompat: a greedy algorithm adding compatible splits or clades in order of their support to a majority consensus tree.
- consensusNet (Holland et al. 2004): extending the majority consensus to allow adding splits with a threshold smaller than 0.5.
- maxCladeCred: each internal split or clade gets a score as the fraction of seeing this split/ clade in the sample. The tree with the highest product of the scores presents the maximum clade credibility tree. So it is a tree from the sample, which is not guaranteed for the other consensus trees. However there might be several trees with the same score.

Assigning edge length

These Consensus trees do not have edge length associated with them. The exception is maximum clade credibility tree as it is one tree from the sample. Generally there are two options to assign edge lengths to a tree.

- We can extract and summarize the edge length or the distance from the root for rooted trees for all the bipartitions (edges) or clades (nodes) which are shared of the consensus tree and the sample of trees. For unrooted trees the functions `add_edgelength` computes the assign them derived from the splits or found in the sampled trees. If the tree is derive by any of the consensus tree methods there should be at least one split

/ clade also in the sample. If we choose an arbitrary tree this is not guaranteed.

- We can use an alignment when available as compute edge length for a given tree using a distance based method (`nnls.tree`), maximum parsimony (`acctran`) or maximum likelihood (`optim.pml`).

Figure 7 shows different consensus trees.

```
strict_consensus <- consensus(trees, rooted=TRUE) |>
    add_edge_length(trees) |> plotBS(cex=0.5, main="a)")
majority_consensus <- consensus(trees, p=.5,
                                rooted=TRUE) |>
    add_edge_length(trees) |> plotBS(cex=0.5, main="b)")
all_compat <- allCompat(trees, rooted=TRUE) |>
    add_edge_length(trees) |> plotBS(cex=0.5, main="c)")
max_clade_cred <- maxCladeCred(trees) |>
    plotBS(cex=0.5, main="d)")
```

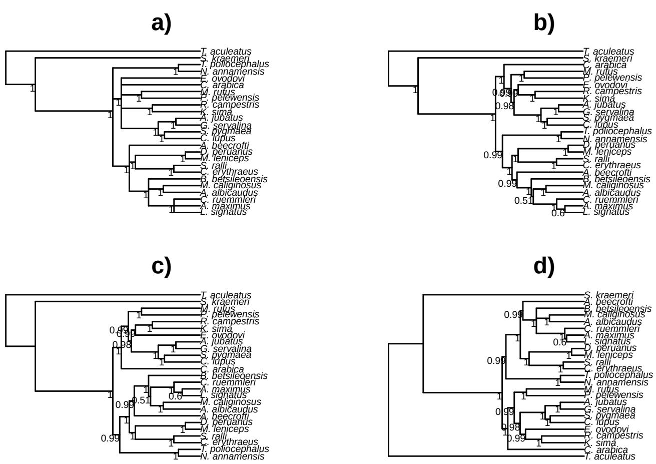

Figure 7: Different consensus trees. a) Strict consensus, b) Majority Consensus, c) Majority consensus tree with compatible splits, d) Maximum clade credibility tree, e) Consensus network. See main text for additional details.

Relationship of *phangorn* to other packages and software

The *phangorn* package has grown to become (along with *ape*(Emmanuel Paradis and Schliep 2019), *phytools* (Liam J. Revell 2024) and *geiger*) among the most important core packages for phylogenetic analysis in the R environment. Nowadays around 50 packages on CRAN or Bioconductor depend on the *phangorn* package. As of the time of writing, the original publication describing *phangorn* (K. P. Schliep 2011) had been cited more than 4,000 times on *Google Scholar* and continues to be cited over 400 times per year.

From the start *phangorn* has been relying heavily on object classes and methods of the core R phylogenetics package, *ape* (E. Paradis, Claude, and Strimmer 2004; Emmanuel Paradis and Schliep 2019). *phangorn* has its own class *phyDat* for storing alignments, but there are many convenience functions to easily transform to the classes `AAbin` and `DNAbin` (*ape*) or to `data.frames` for categorical data. Several frequently used functions in *phangorn* (`bab`, `pml_bb`, `dist.ml`, `pratchet`) accept the classes `AAbin` and `DNAbin` from *ape* directly as input. Furthermore alignments can exchanged with the package *Biostrings* (Pagès et al. 2022) and connect to the Bioconductor world.

Apart from the *ape* package *phangorn* currently depends on a number of other packages. First of all *phangorn* depends on many of the R core packages (R Core Team 2020). Furthermore on the packages *checkmate* (Lang 2017), *digest* (Dirk Eddelbuettel with contributions by Antoine Lucas et al. 2022), *fastmatch* (Urbanek 2021), *generics* (Wickham, Kuhn, and Vaughan 2022), *igraph* (Csardi and Nepusz 2006), *Matrix* (Bates, Maechler, and Jagan 2023), *osqp* (Stellato et al. 2024), *future.apply* (Bengtsson 2021) and *Rcpp* (Eddelbuettel and Balamuta 2018; Eddelbuettel and François 2011; Eddelbuettel 2013). This relationship evolves with the package development and might change.

Ancestral reconstructions from *iqtree* or networks from *Splitstree* can be imported and visualized.

Conclusions

More than a decade has passed since the original article describing *phangorn* was published (K. P. Schliep 2011). Since that time, the *phangorn* package has both evolved into one of the core function libraries of the R phylogenetics ecosystem, and expanded in size an scope. We decided the literature reference for *phangorn* was sorely in need of updating. In creating one, we were determined to make something that could serve as more than a placeholder to capture citations of the *phangorn* package. We hope that this will help guide new *phangorn* users towards interesting analytical tools, as well as perhaps inspire experienced *phangorn* and R phylogenetics researchers to generate new types of questions and data that will in turn help motivate and participate in the continued development of the *phangorn* package into the future.

Software and data availability

The *phangorn* is free and open source, and can be downloaded from its CRAN (https://CRAN.R-project.org/package=phangorn) or the development version from GitHub (https://github.com/KlausVigo/phangorn) pages. Binaries for the development version are kindly supplied on the r-universe pages (https://klausvigo.r-universe.dev/phangorn#).

This article was written in quarto and developed with the help of both the posit Rstudio IDE (RStudio Team 2023).

The underlying markdown code and data files necessary to reproduce the analyses of this article are available at https://github.com/KlausVigo/phangorn-v3/.

Competing interests

No competing interest is declared.

Author contributions statement

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